

AI-Powered Classical Arabic & Quranic Learning Platform

System Modules — Detailed Documentation

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2 Modules | 8 Categories

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■ Auth & User Management

01 User Registration & Login

Secure account creation and authentication system for all platform users.

Key Features:

- Email-based registration with OTP verification
- JWT-based stateless authentication with bcrypt password hashing
- Social login support (Google OAuth 2.0)
- Password reset via email link
- Session expiry and refresh token rotation

Technologies: *FastAPI, PostgreSQL, bcrypt, JWT, Google OAuth 2.0*

02 User Profile & Settings

Personalized user profiles with learning preferences and account customization.

Key Features:

- Profile picture upload and display name editing
- Interface language preference (English / Urdu)
- Notification preferences (email, push, in-app)
- Daily study goal configuration (10 / 20 / 30 mins)
- Account deletion and data export (GDPR-friendly)

Technologies: *FastAPI, PostgreSQL, AWS S3 / local file storage*

03 Role Management

Granular access control separating student, content editor, and administrator roles.

Key Features:

- Three roles: Student, Content Editor, Super Admin
- Role-based API endpoint guards via FastAPI dependencies
- Admin can promote/demote users between roles
- Content editors can manage lessons and vocabulary sets
- Audit log for all admin actions

Technologies: *FastAPI RBAC middleware, PostgreSQL roles table*

04 Session & Device Management

Full visibility and control over active login sessions across devices.

Key Features:

- List all active sessions with device type and last seen
- Revoke individual sessions or log out all devices
- Suspicious login alerts via email
- Auto-logout after configurable inactivity period
- Mobile and web session differentiation

Technologies: *Redis (session store), JWT refresh tokens, FastAPI*

■ Quran Browser & Content

05 Quran Browser

Full Quranic text navigation with Arabic script, translations, and verse-level controls.

Key Features:

- Browse by Surah (114 chapters), Ayah, or Juz (30 parts)
- Multiple English translations (Sahih International, Yusuf Ali)
- Arabic text with full diacritics (tashkeel / harakat)
- Continuous scroll and page-by-page reading modes
- Font size control and night mode

Technologies: *quran.com API, React.js, React Native*

06 Word-by-Word Translation

Tap any Arabic word in any verse for instant meaning, root, and grammar info.

Key Features:

- Tap/click any word to open a details panel
- Displays word meaning, transliteration, and grammar type
- Links directly to the Sarf Engine for full morphological analysis
- Highlights all occurrences of that word across the Quran
- Data sourced from corpus.quran.com annotation dataset

Technologies: *corpus.quran.com dataset, quran.com API, React*

07 Ayah Bookmarking & Notes

Save and annotate important verses with personal study notes.

Key Features:

- Bookmark any ayah with one tap
- Add personal text notes to any verse
- Organize bookmarks into named collections (e.g. Tawheed, Dua)
- Export bookmarks as PDF or plain text
- Sync across web and mobile via API

Technologies: *PostgreSQL, FastAPI, React / React Native*

08 Root Frequency Explorer

Discover and explore the most frequently occurring Arabic roots in the Quran.

Key Features:

- Ranked list of top 500 roots by Quranic frequency
- Filter by frequency range, part of speech, or chapter
- Click any root to see all its Quranic appearances
- Color-coded frequency heatmap per Surah
- Useful for vocabulary prioritization in learning path

Technologies: *corpus.quran.com dataset, PostgreSQL, Chart.js*

09 Thematic Vocabulary Sets

Pre-curated vocabulary collections grouped by Quranic themes and topics.

Key Features:

- Themes: Prayer, Creation, Prophets, Judgment Day, Ethics, Nature
- Each set contains 15-30 key words with roots and meanings
- Directly launchable as SRS flashcard decks
- Admin CMS for creating and editing theme sets
- User can subscribe to themes for inclusion in daily reviews

Technologies: *PostgreSQL, FastAPI, React / React Native*

■ Sarf / Morphology Modules

10 Root Analyzer

Core morphological engine that decomposes any Arabic word into its root and pattern.

Key Features:

- Input any Arabic word (with or without diacritics)
- Returns 3-letter root (■■■■), verb pattern (■■■■), and morphological class
- Displays grammatical features: tense, voice, gender, number, person
- Lists all Quranic occurrences of the identified root
- Powered by CAMEL Tools morphological analyzer

Technologies: CAMEL Tools (CMU), FastAPI NLP endpoint, React UI

11 Verb Conjugation Trainer

Interactive training module for all 15 classical Arabic verb forms (■■■■■■).

Key Features:

- Displays full conjugation tables for any root across all 15 forms
- Drag-and-drop conjugation practice exercises
- Fill-in-the-blank verb form completion challenges
- Progress tracked per verb form (■■■■) individually
- Audio pronunciation for each conjugated form (TTS)

Technologies: CAMEL Tools, FastAPI, React / React Native

12 Word Family Explorer

Visual explorer showing all derivatives and word families from a single Arabic root.

Key Features:

- Enter root letters to see all derived words in Arabic
- Groups derivatives by type: nouns, verbs, adjectives, participles
- Example: root ■-■-■ produces ■■■■■ ■■■■■ ■■■■■ ■■■■■ etc.
- Each word links to its Quranic occurrences
- Interactive tree visualization of the word family

Technologies: corpus.quran.com dataset, CAMEL Tools, D3.js for tree viz

14 Nahw Parser (Grammar Annotator)

Key Features:

- Technologies:** *Farasa NLP (KACST), CAMEL Tools, FastAPI, React*

Key Features:

- Technologies:** *FastAPI (quiz engine), PostgreSQL, React / React Native*

Key Features:

- Technologies:** *FastAPI, PostgreSQL, corpus.quran.com data, React*

17 Grammar Rule Library

Searchable reference encyclopedia of classical Arabic grammar rules with Quranic proofs.

Key Features:

- 200+ grammar rules categorized by topic (nominal, verbal, particle sentences)
- Each rule illustrated with a Quranic example verse
- Search by keyword, grammar term, or browse by category
- Bookmarkable rules for personal reference
- Linked to i'rab practice exercises for each rule

Technologies: PostgreSQL (rule database), FastAPI, React search UI

■ Vocabulary & Flashcards

18 Flashcard SRS System

Science-based spaced repetition flashcard engine for long-term Quranic vocabulary retention.

Key Features:

- SM-2 algorithm schedules each card's next review based on recall rating
- Card shows: Arabic word → root → meaning → Quranic verse example
- User rates recall: Again / Hard / Good / Easy (1-4 scale)
- Daily review queue shows only due cards, preventing overload
- Mastery levels: New → Learning → Review → Mastered

Technologies: FastAPI, PostgreSQL (SRS scheduling), React / React Native

19 Vocabulary Statistics Dashboard

Visual analytics dashboard showing learner's vocabulary growth and retention trends.

Key Features:

- Total words learned, mastered, and due for review
- Daily/weekly retention rate chart
- Heatmap calendar of study activity (GitHub-style)
- Weak words list — cards with lowest recall scores
- Forecast: projected vocabulary size over next 30 days at current pace

Technologies: FastAPI, PostgreSQL, Chart.js, React

20 Custom Deck Builder

Tool for learners to create, share, and study personalized Quranic vocabulary decks.

Key Features:

- Create a named deck and add words from any Quranic search
- Import words from a specific Surah automatically
- Share decks publicly with other users via shareable link
- Clone and modify community-shared decks
- Decks integrate seamlessly into the SRS review queue

Technologies: PostgreSQL, FastAPI, React / React Native

■ AI & Personalization

21 AI Tutor Chatbot

Grammar-aware conversational AI tutor powered by Gemini 2.0 Flash, specialized in Classical Arabic.

Key Features:

- Answers grammar and morphology questions in English with Arabic examples
- Explains any Quranic verse with word-by-word grammatical breakdown on request
- Supports follow-up questions with full conversation context
- System prompt engineered with Classical Arabic pedagogy and Nahw/Sarf terminology
- Tutor can quiz the user interactively (e.g. 'What is the root of ■■■■■■■■?')

Technologies: Gemini 2.0 Flash API, FastAPI, WebSocket / REST, React

22 Adaptive Learning Path

AI-generated personalized curriculum based on learner's assessed entry level.

Key Features:

- Onboarding quiz (20 questions) assesses vocabulary, grammar, and morphology level
- System generates a sequenced curriculum: Beginner → Intermediate → Advanced
- Lessons unlock progressively as prior competencies are demonstrated
- Path includes mix of vocabulary, Sarf, and Nahw modules per week
- User can override and explore any module freely

Technologies: FastAPI (curriculum engine), PostgreSQL, React UI

23 Weak Area Detection

Automated tracking of performance across all modules to identify and remediate weak areas.

Key Features:

- Aggregates error data from i'rab exercises, SRS ratings, and quizzes
- Identifies specific struggling concepts (e.g. 'broken plurals', 'Form II verbs')
- Automatically resurfaces weak concepts in daily practice queue
- Weekly weak area summary report shown on dashboard
- AI Tutor is context-aware of flagged weak areas and proactively addresses them

Technologies: FastAPI analytics engine, PostgreSQL, Gemini API integration

24 AI-Generated Quizzes

Dynamic quiz generation via Gemini API, tailored to each learner's current level and weak areas.

Key Features:

- Gemini generates novel grammar and vocabulary questions — no fixed question bank
- Quiz types: MCQ, fill-in-the-blank, sentence translation, root identification
- Difficulty auto-calibrated based on learner's performance history
- Answers explained with grammar rule references and Quranic examples
- Quiz results feed back into weak area detection module

Technologies: Gemini 2.0 Flash API, FastAPI, PostgreSQL, React

■ Gamification & Engagement

25 Daily Challenge Module

One fresh grammar or vocabulary challenge per day to build consistent learning habits.

Key Features:

- One AI-generated challenge delivered daily (grammar puzzle or vocab set)
- Challenges rotate across module types: Sarf, Nahw, vocabulary
- Time-limited challenge with countdown timer for added engagement
- Global daily challenge — same challenge for all users enables discussion
- Bonus XP awarded for completing challenge within first hour of release

Technologies: Gemini API (generation), FastAPI cron job, Redis, React Native push

26 Streak & Habit Tracker

Daily login and study streak system that reinforces consistent learning habits.

Key Features:

- Streak increments when user completes at least one study activity per day
- Streak freeze item available (earned through achievements) to protect streaks
- Weekly goal setting: e.g. study 5 out of 7 days
- Visual streak calendar on profile page
- Push notification reminder 1 hour before midnight if streak at risk

Technologies: FastAPI, PostgreSQL, Redis (streak state), FCM push notifications

27 Achievement & Badge System

Milestone-based badge system celebrating learner progress across all platform modules.

Key Features:

- 30+ badges across categories: Vocabulary, Grammar, Streaks, Exploration
- Example badges: 'Root Hunter' (analyzed 50 roots), 'Nahw Ninja' (100 correct i'rab)
- Badges visible on user profile and shareable as images
- Rare badges for major milestones (e.g. completed all 32 modules)
- Badge unlock animation with motivational message

Technologies: PostgreSQL (badge definitions), FastAPI event triggers, React Native

28 Leaderboard

Weekly and all-time leaderboards to foster friendly competition among learners.

Key Features:

- Weekly leaderboard resets every Monday — fresh competition each week
- XP scoring: earned from flashcard reviews, quiz completions, daily challenges
- Global leaderboard and friends-only leaderboard
- Top 3 winners highlighted with special weekly crown badges
- Opt-out option for users who prefer private learning

Technologies: *Redis (leaderboard sorted sets), FastAPI, PostgreSQL, React*

■ Admin & System Modules

29 Admin Dashboard

Central command panel for platform administrators to monitor and manage the system.

Key Features:

- Real-time user statistics: total registered, daily active, new signups
- Module usage analytics: most used features, drop-off points
- User management: search, ban, promote, view activity logs
- System health monitoring: API response times, DB query performance
- Content moderation queue for user-submitted notes and decks

Technologies: *FastAPI admin endpoints, PostgreSQL, Chart.js, React admin panel*

30 Content Management System (CMS)

Backend CMS for managing lessons, grammar rules, vocabulary sets, and thematic content.

Key Features:

- Rich text editor for creating and editing grammar lessons
- Vocabulary set creation with Arabic input support and root validation
- Lesson sequencing: drag-and-drop ordering within learning path
- Version history and rollback for edited content
- Publish / draft / archived states for all content types

Technologies: *FastAPI, PostgreSQL, React admin with Arabic-capable rich text editor*

31 Search Module

Full-text Arabic and English search across the entire platform — Quran, grammar, vocabulary.

Key Features:

- Search Quranic text in Arabic or by English keyword
- Search grammar rule library by concept name or Arabic term
- Search vocabulary database by English meaning or Arabic word
- Autocomplete suggestions as user types
- Results ranked by relevance with module-type filtering

Technologies: *PostgreSQL full-text search (pg_trgm), FastAPI, React search UI*

32 Push Notification & Reminder System

Smart notification engine delivering personalized reminders to maintain study consistency.

Key Features:

- Daily SRS review reminder when cards are due
- Streak danger alert: notifies 1 hour before midnight if streak at risk
- Weekly progress summary via email
- Daily challenge release notification
- User controls notification type and timing in profile settings

Technologies: *Firebase Cloud Messaging (FCM), FastAPI background tasks, Redis queue*